Rent Inflation and Monetary Policy

Sergi Barcons[1] Gonzalo Paz-Pardo[2]

[1]Toulouse School of Economics

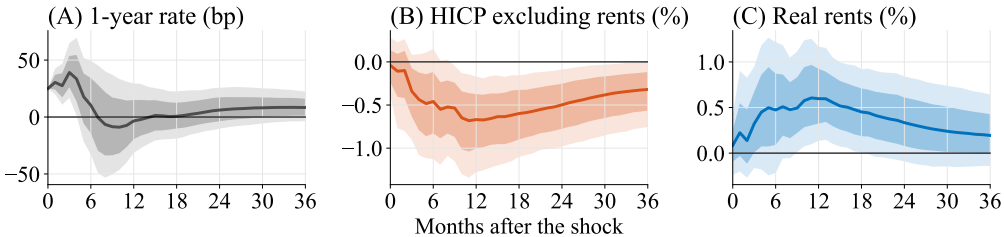
[2]European Central Bank

November 2, 2026

The views expressed here are those of the authors and do not necessarily reflect those of the European Central Bank or the Eurosystem.

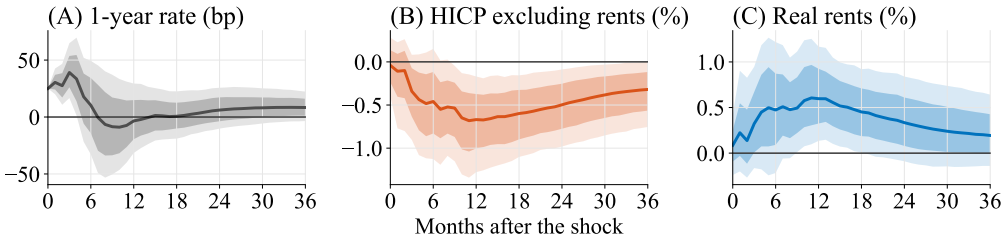
Motivation

- Unlike other prices, rents do not fall when policy tightens: **real rents rise**



Motivation

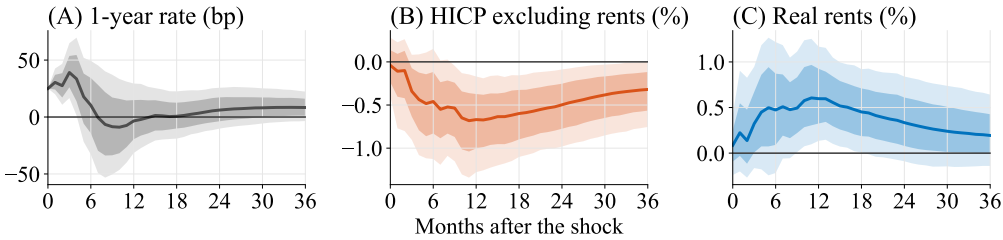
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- Rents are part of the **inflation target**

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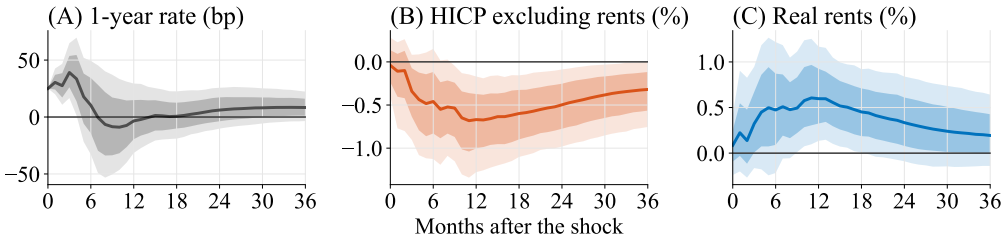


- Rents are part of the **inflation target**

  → *relevance*: **a third** of a renter's income, and renters earn **a quarter less**

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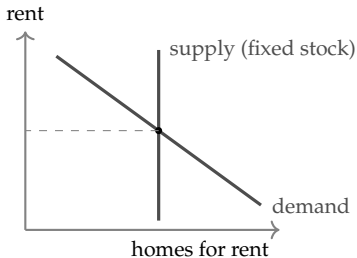
- Rents are part of the **inflation target**
  → *relevance*: **a third** of a renter's income, and renters earn **a quarter less**

Q: Why do **real rents rise** when monetary policy tightens?
What should **central banks** do about it?

▸ VAR details

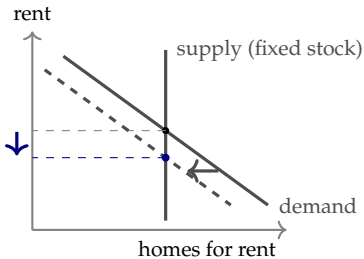
Why Rents Rise

Textbook *frictionless* economy



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Textbook *frictionless* economy



higher rates $\Rightarrow$ people spend less

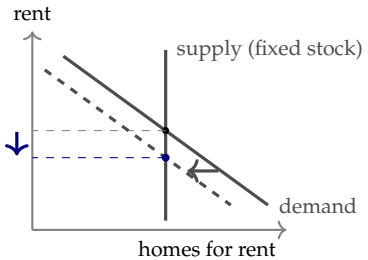
$\Rightarrow$ rental demand $\downarrow$ $\Rightarrow$ **rents fall**

▸ Textbook model

standard NK mechanism

Why Rents Rise

Textbook *frictionless* economy

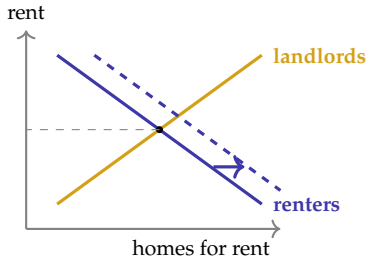


higher rates ⇒ people spend less
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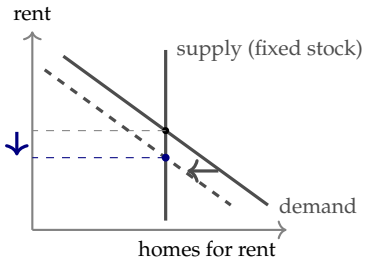
Our economy



More renters: buyers priced out of owning

Why Rents Rise

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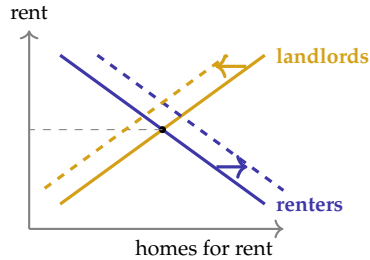


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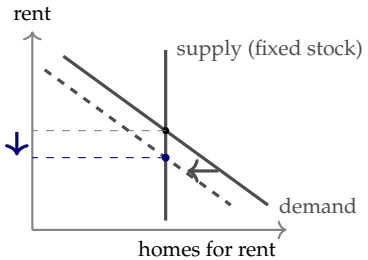


More renters: buyers priced out of owning
Fewer homes for rent: holding costs rise

▸ Who are the landlords?

Why Rents Rise

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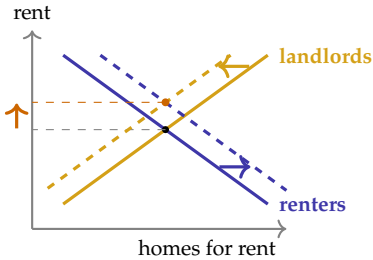


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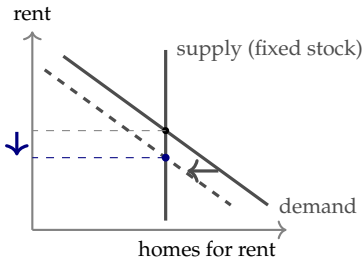
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new here: landlords weigh rents against interest rates

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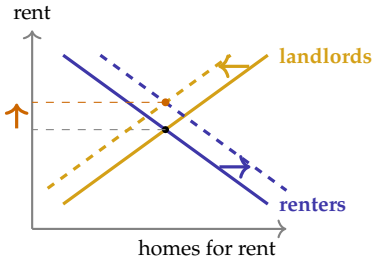


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Our economy



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new here: landlords weigh rents against interest rates

This paper: which force dominates, and what should policy do? ▸ Who are the landlords?

This Paper

- A **HANK model** where households rent, own, or become landlords
  $\rightarrow$ landlords are households: rental supply is a portfolio choice

- Three steps:
  1. **Rent channel:** how monetary policy moves rents, via landlords or buyers
  2. **Rent in the target:** what happens when rents enter the target, as in the HICP
  3. **Optimal response:** how policy should respond to rent inflation

Results

Rent channel

1. A tightening raises rents while house prices fall
2. **Landlords**, not **would-be buyers**, drive the rent rise
   → about 65% of the pressure from higher rates: returns on savings, mortgage costs

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Rent in the target

3. Putting rents in the target deepens the contraction
4. The rule fights rent inflation it created: **self-defeating**
   → rents paid lag, so it outlasts the shock → lowers welfare under all shocks considered

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Optimal response

5. Neither tighten against rent inflation nor look through it: **ease**
6. Why: mostly redistribution; under cost-push, efficiency too
   → rent relief redistributes, but also insurance benefits
7. **Full Ramsey plan**: most of the gain is a time-0 redistributive surprise

Related Literature

Housing and tenure choice in general equilibrium: Sommer–Sullivan–Verbrugge (2013), Kaplan–Mitman–Violante (2020), Castellanos–Hannon–Paz-Pardo (2024), Greenwald–Guren (2025)

Contribution **1.** Monetary transmission via the rental market: why rents rise when policy tightens

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Contributions
2. Endogenous tenure transitions + household landlords
3. Optimal sign of the response to rent inflation: ease, not exclude or down-weight

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HANK and optimal monetary policy: Kaplan–Moll–Violante (2018), Bhandari et al. (2021), Acharya–Challe–Dogra (2023), Dávila–Schaab (2023, 2025)

Contribution **4.** Rental market as new source of efficiency + redistribution motives; Ramsey policy

Outline

Why do rents rise when policy tightens?

1. Model
2. Calibration
3. Rent channel

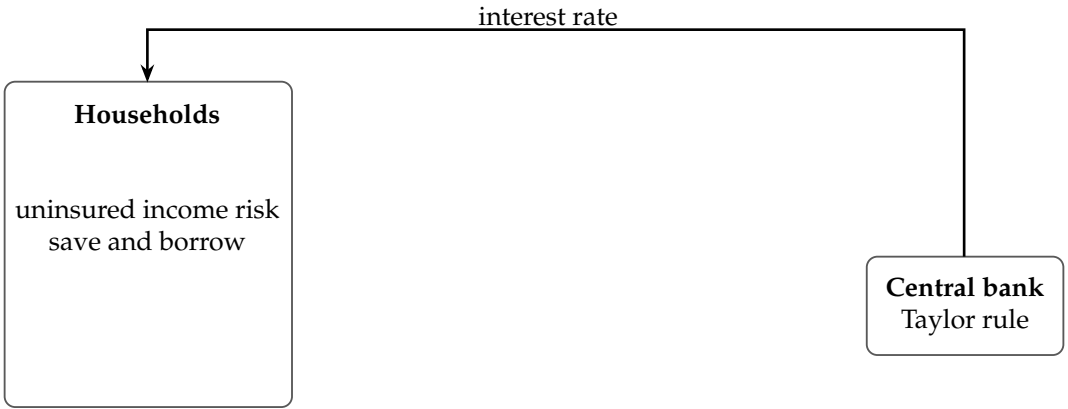
What should central banks do?

4. Rent in the target
5. Optimal response

Model

Model Overview

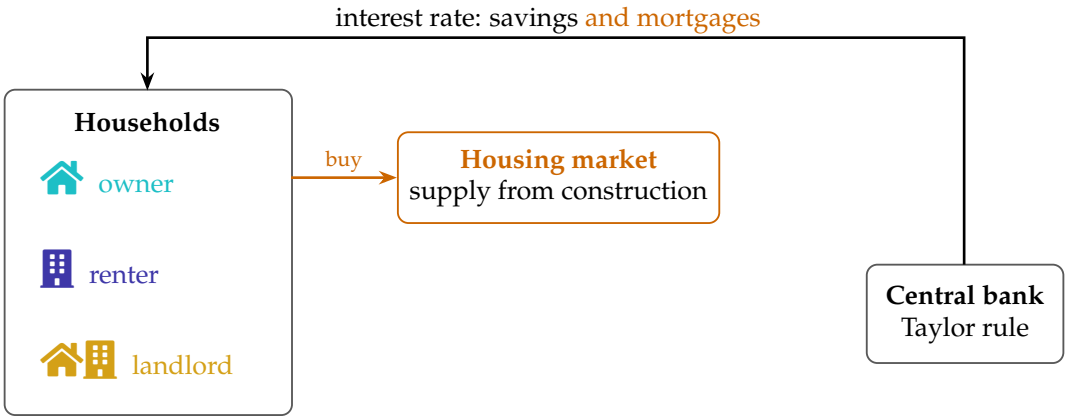
Standard HANK



Standard blocks: sticky wages, government debt

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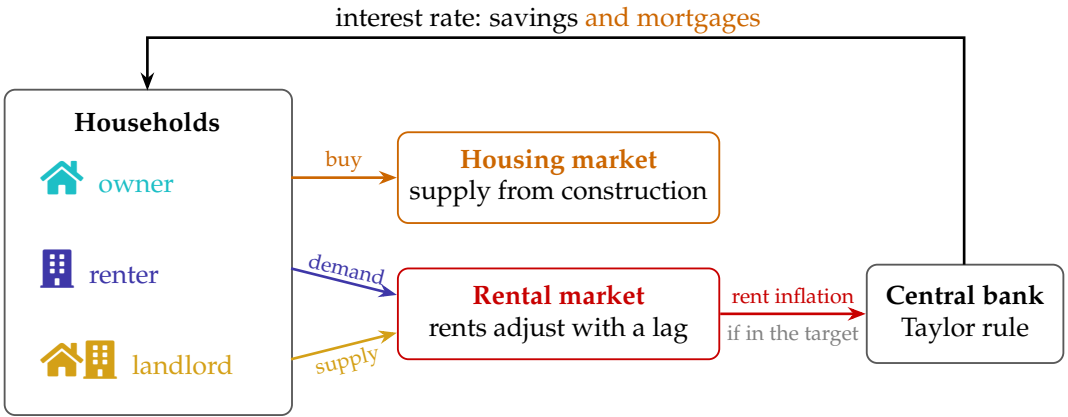
Standard HANK + housing market



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Model Overview

Standard HANK + housing market + rental market



Standard blocks: sticky wages, government debt

Households: Renters, Owners, Landlords

▸ Recursive problem

Preferences

$$\mathcal{U}(c,h,n) = \frac{\left(c^{f(h)}\right)^{1-\gamma}}{1-\gamma} - \psi_n \frac{n^{1+\eta}}{1+\eta}$$



Renter ($h = 0$)

→ rents a small unit
→ pays rent $\widehat{p}^r$
→ cannot borrow
→ saves in a risk-free deposit at r

Budget constraint ▸ Housing costs

$$c + a' + \psi_t(h,h') \leq \left(1 + r_t + \kappa \mathbb{1}_{a<0}\right) a + (1-\tau_y)\, w_t\, y\, n + T_t - \widehat{p}^r_t$$

Borrowing limit

$$a' \geq 0$$

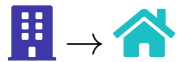
State: wealth a, housing h, AR(1) productivity y + **taste shocks** ⇒ logit tenure choice

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A renter who buys

→ buys a larger home: price + transaction tax

→ mortgage: up to a share λ_{LTV} of the home's value and λ_{LTI} times income

Budget constraint Housing costs

$$c + a' + \psi_t(h,h') \;\leq\; \left(1 + r_t + \kappa\, \mathbb{1}_{a<0}\right) a + (1-\tau_y)\, w_t\, y\, n + T_t - \widehat{p}_t^{\,r}$$

Borrowing limit $p^h\, q(\cdot)$: value of the housing bought

$$a' \;\geq\; -\lambda_{\text{LTV}}\, p_t^h\, q(h,h') \;\;\&\;\; -\lambda_{\text{LTI}}\, w_t\, y\, n$$

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Owner ($h = 1$)

$\rightarrow$ lives in a larger home: $f(h) > 1$

$\rightarrow$ pays house depreciation

$\rightarrow$ mortgage pays $r + \kappa$

$\rightarrow$ amortizes at rate m

Budget constraint

Housing costs

$$c + a' + \psi_t(h,h') \quad \leq \quad \left(1 + r_t + \boldsymbol{\kappa}\,\mathbb{1}_{a<0}\right) a + (1-\tau_y)\, w_t\, y\, n + T_t - \widehat{p}_t^r$$

Borrowing limit

$$a' \;\geq\; \boldsymbol{(1-m)\,a}$$

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$$\mathcal{U}(c,h,n) = \frac{\left(cf(h)\right)^{1-\gamma}}{1-\gamma} - \psi_n \frac{n^{1+\eta}}{1+\eta}$$



Landlord ($h = 2$) Rental supply

$\rightarrow$ owns two homes, lives in the larger one

$\rightarrow$ rents the small one out: receives rent $\widehat{p}^r$

Budget constraint Housing costs

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Rental Market and Sticky Leases

Leases

→ contract: match between one **renter** & one **landlord**

→ rent set at signing: p^r_{t-s} *(in real terms)*

→ new lease: *(exogenous)* renegotiation or tenure change

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Rent paid

→ indexed to goods inflation π^c

→ after J quarters: long-term tenants ➜ average of old signing rents

→ rent index = average rent paid ▸ Average rent

⇒ **Rents paid catch up as leases turn over**

Remaining Model Details

Wages and goods sticky wages: unions ration hours, pay a Rotemberg adj. cost ▸ Details

goods: $Y_t^c = A^c N_t$, competitive firm $\Rightarrow$ $w_t = A^c$, $\pi_t^w = \pi_t^c$

Construction new homes from land permits and goods

Central bank $i_t = r_{ss} + \phi_\pi \pi_t^{\text{hicp}} + \varepsilon_t^m, \quad \pi_t^{\text{hicp}} = (1-\omega_r)\,\pi_t^c + \omega_r\,\pi_t^{\text{rent}}$ ▸ Other rules ▸ Spain?

$\rightarrow \pi^{\text{rent}}$: inflation of the rent index $\rightarrow \omega_r = 0$: goods only $\rightarrow \omega_r > 0$: rents in the target

Government debt absorbs deficits; the lump-sum transfer responds to debt

Equilibrium prices clear housing, rental (**renters** = **landlords**), bonds and land

▸ Equilibrium

Calibration

Calibration to Spain

Why Spain → most outstanding mortgages are adjustable-rate, as in the model
→ landlords are households: they own 92% of private rentals
institutional investors: 34% in Germany, 23% in France

Set externally ▸ All parameters

Leases $\varphi = 0.99 \Rightarrow$ 17% signed a year
data $\in [14\%, 20\%]$

Mortgages λ_{LTV}: 80% m: 90% in 30 years
λ_{LTI}: 6× annual income
κ: 1.38 pp spread

Policy $\phi_\pi = 1.5$

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Set internally, jointly

house value / income	A^h
construction / GDP	δ_h
renter share	θ
renter → owner flow	τ_h
landlord → owner flow	σ_ε
2% real rate	β
public spending / GDP	τ_y

Model Fit

Targeted	**Model**	**Spain**
house value / annual income	5.25	5.25
residential construction / GDP	6.9%	6.9%
renter share	17.0%	17.0%
public spending / GDP	19.0%	19.0%

Model Fit

Targeted	Model	Spain
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renter share	17.0%	17.0%
public spending / GDP	19.0%	19.0%
Untargeted		
rent burden (renters)	30.4%	35.0%
owners + landlords	83.0%	82.9%
owner-occupiers	66.0%	72.9%
landlords	17.0%	10.0%
net worth / annual income	6.58	7.33

$\rightarrow$ model: one home per landlord $\rightarrow$ data: several

Model Fit

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Tenure transitions, quarterly * targeted

	next quarter: renter	owner	landlord
renter	98.8 98.8	1.2* 1.2	0.0 0.0
owner	0.3 0.1	99.0 99.3	0.7 0.6
landlord	0.0 0.0	2.6* 2.6	97.4 97.4

model, %
data, %

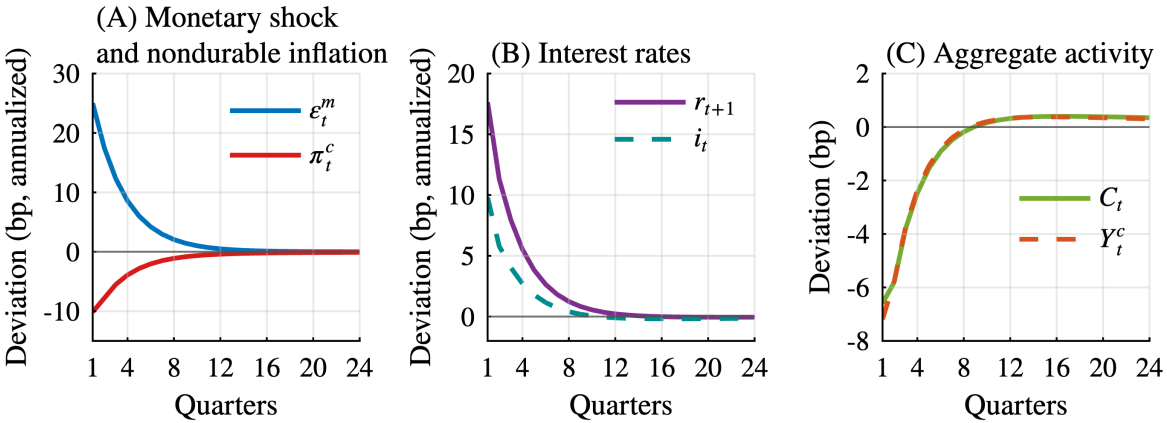
EFF panel 2014–2022, converted to quarterly

▸ More moments

Rent Channel

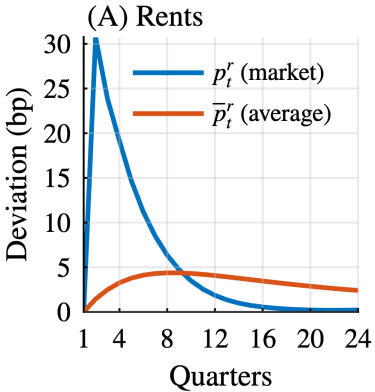
Monetary Shock: Aggregates

Shock +25 bp to the Taylor rule (annualized), persistence 0.7

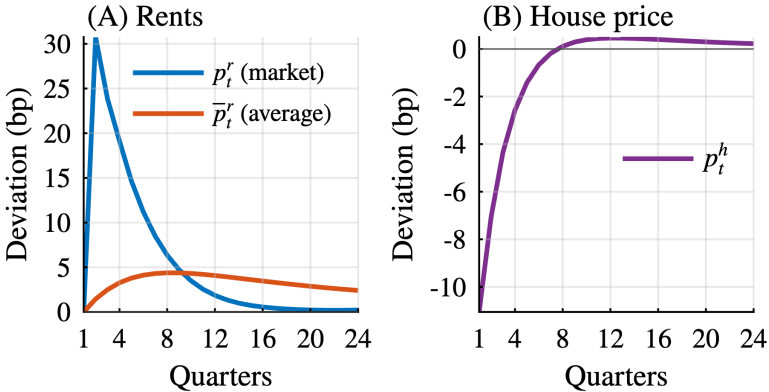


A standard HANK response real rate +71 bp per 100 bp shock (Kaplan et al. 2018: $\approx$ 60 bp)

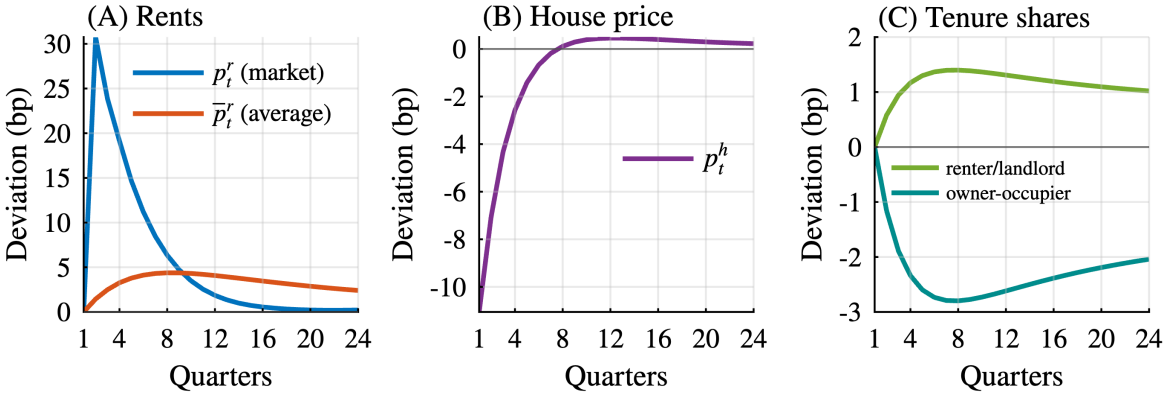
Monetary Shock: Rents and House Prices



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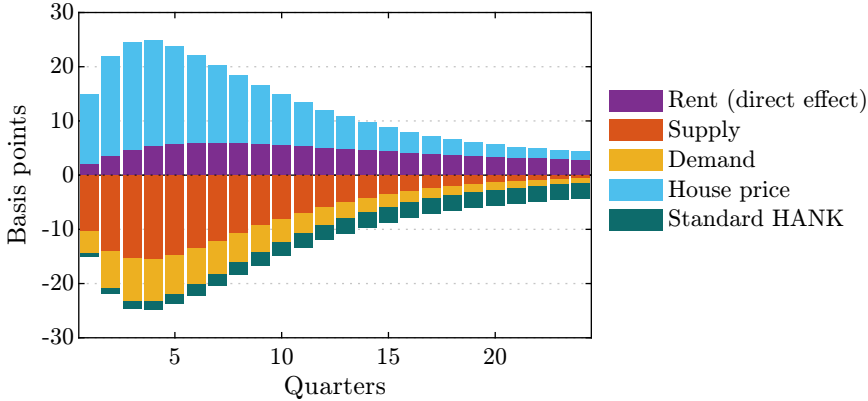


As in the data: homeownership falls, rent-to-price ratio rises

→ more renters ⇒ rents rise until enough households become landlords ▸ Model vs data

Rent Decomposition

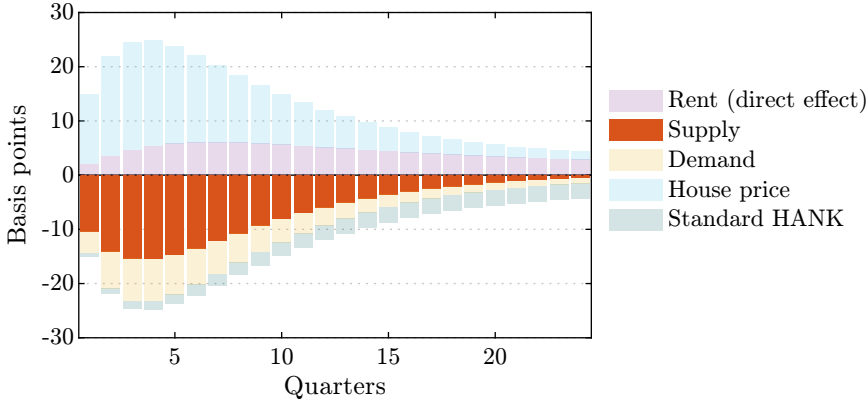
Bars: excess rental supply
below zero ⇒ pushes rents up



Partial equilibrium: each color moves one price alone, the others stay at SS
→ its contribution to the rent rise

Rent Decomposition

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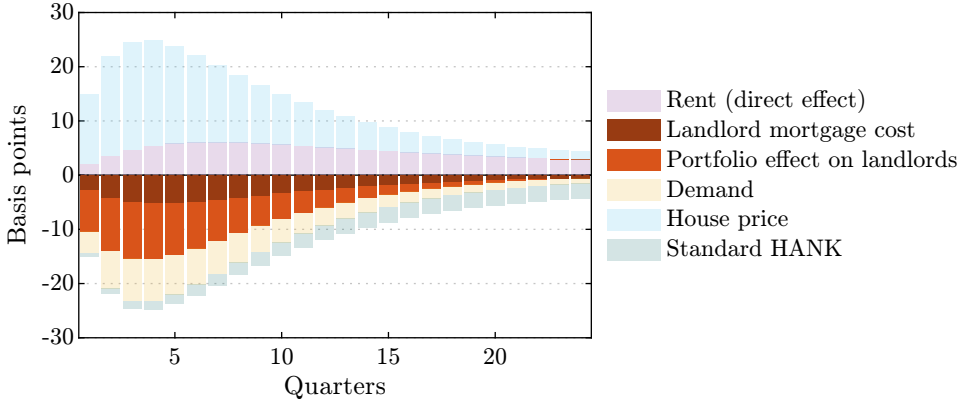


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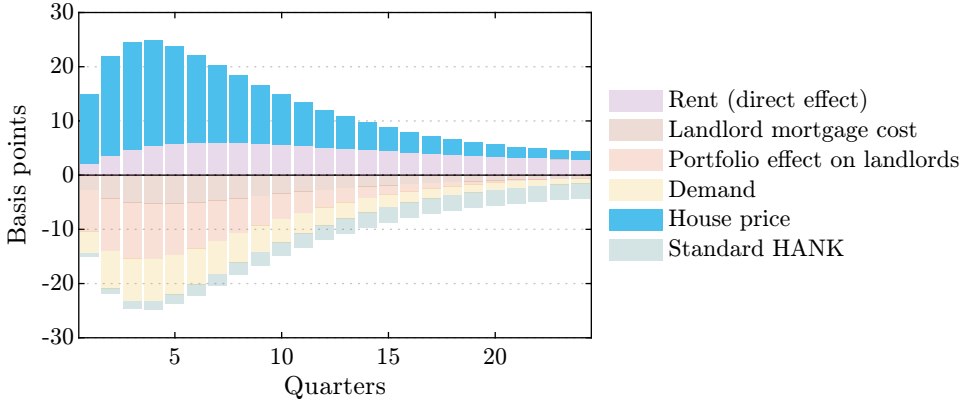


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Rent Decomposition

Bars: excess rental supply
below zero ⇒ pushes rents up



Landlords, not would-be buyers, drive the rent rise

→ ≈ 65% of the pressure from higher rates: savings returns, landlord mortgage costs

→ falling house prices offset ≈ two-thirds

Taking Stock

1. **HANK with rental supply**: households rent, own, or become landlords
2. Tightening $\rightarrow$ **real rents** $\uparrow$, house prices $\downarrow$, as in the data
3. **Mostly supply**: landlords, $\approx 65\%$ of the rate pressure

$\Rightarrow$ **What should central banks do about it?**

What Should Central Banks Do?

Central Bank Response

1. **Rent in the target**: HICP-style rule backfires — ▸ Rent in the target
2. **Optimal response**: **ease**, rent inflation lowers the rate — ▸ Optimal rule
3. **Ramsey**: gains mostly a time-0 redistributive surprise — ▸ Ramsey
4. Rent relief is progressive — ▸ Who gains
5. Mostly redistribution; cost-push: efficiency too — ▸ Efficiency *vs* redistribution
6. Not just a less hawkish rule — ▸ Less hawkish?

Conclusion

1. **Rent channel:** a tightening raises rents while house prices fall
   → driven by landlords: better returns on savings, higher mortgage costs

2. **Rent in the target:** the HICP-style rule backfires
   → deeper contraction, and welfare losses under all shocks considered

3. **Optimal response:** rent inflation should lower the policy rate
   → rent relief for renters, lower mortgage costs for owners

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Tighten against rent inflation, look through it, or ease? **Ease.**

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Tighten against rent inflation, look through it, or ease? **Ease.**

Thank you for your attention!

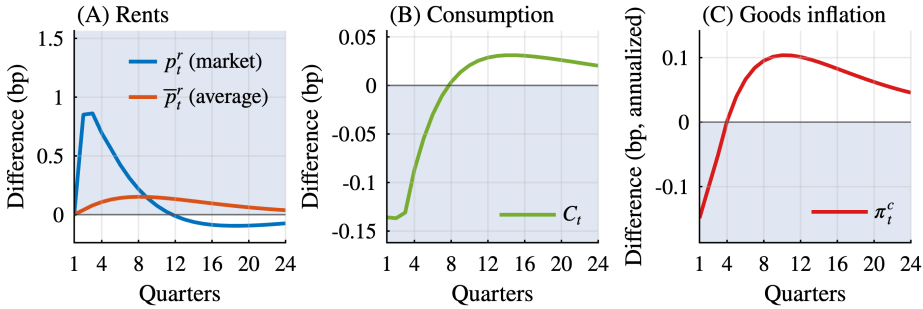
Rent in the Target

Rent in the Target: Policy Feedback

Goods only $i_t = r_{ss} + \phi_\pi \, \pi_t^c + \varepsilon_t^m$

HICP-style $i_t = r_{ss} + \phi_\pi \, \pi_t^{\text{hicp}} + \varepsilon_t^m, \quad \pi_t^{\text{hicp}} = (1 - \omega_r) \, \pi_t^c + \omega_r \, \pi_t^{\text{rent}}$

$\omega_r = 0.053$: model's rent share



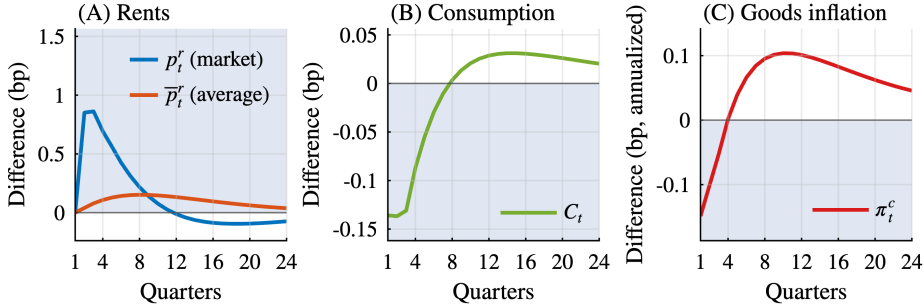
HICP-style minus goods only rule amplifies the tightening

Back

Rent in the Target: Policy Feedback

Goods only $i_t = r_{ss} + \phi_\pi \, \pi_t^c + \varepsilon_t^m$ $\omega_r = 0.053$: model's rent share

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HICP-style minus goods only rule amplifies the tightening

Rents in the target destabilize

+2.7% consumption loss | **+3.4%** rents paid | **+2.8%** π^c volatility | **+2.8%** π^{rent} volatility

for no extra disinflation (0% at two years, −2.7% by year five) ▸ All weights ▸ Owner-occupied housing ◂ Back

Taking Stock

1. **HICP-style rule**: tightening $\rightarrow$ **rents** $\uparrow$ $\rightarrow$ rule tightens more
2. Rents paid lag $\rightarrow$ the loop **outlasts the shock**
3. **Deeper contraction**: the rule fights rent inflation it created

$\Rightarrow$ **How should policy respond to rent inflation?**

Back

Optimal Response

Optimal Simple Rule

Rule $i_t = r_{ss} + \phi_\pi \, \pi_t^c + \phi_r \, \pi_t^{\text{rent}}$ → main exercise: ϕ_π fixed, vary ϕ_r

Welfare

$$\phi_r^* = \arg\max_{\phi_r} \sum_{t \geq 1} \tilde{\beta}_t \Big[\underbrace{\int u(c_{it}, h_{it}) \, di}_{\substack{\text{all households,} \\ \text{equal weights}}} - \underbrace{\nu(N_t)}_{\text{hours}} - \underbrace{\frac{\theta_w}{2} (\pi_t^w)^2}_{\text{wage adjustment}} \Big]$$

Four shocks that raise rents

Preferences	▸ IRFs	impatience, and the rule tightens → fewer buyers → rents rise
Govt spend	▸ IRFs	spending ↑ → the rule tightens → fewer buyers → rents rise
Rental supply	▸ IRFs	landlords' mortgage costs ↑ → fewer homes for rent → rents rise
Cost-push	▸ IRFs	wage markups ↑ → π^c ↑ → the rule tightens → rents rise

Should rent inflation raise ($\phi_r > 0$) or lower ($\phi_r < 0$) the policy rate?

◂ Back

Optimal Weight on Rent Inflation

	demand		supply	
	Preferences	**Govt spend**	**Rental supply**	**Cost-push**
Optimal weight, ϕ_r^*	-0.50	-0.50	-0.18	-0.48
Welfare gain at ϕ_r^*	$+1.24$	$+0.37$	$+0.11$	$+0.09$

→ relative to goods only → CE in bp of permanent consumption → demand shocks: at the bound $\phi_\pi + \phi_r = 1$

Ease: rent inflation should lower the policy rate

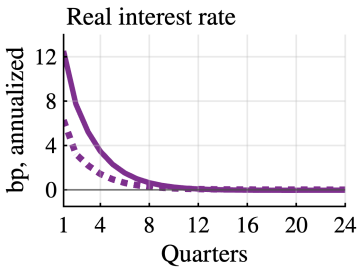
→ **any $\phi_r \in [-0.25, 0]$** raises welfare under all four shocks

→ HICP-style rule: welfare losses under all four ▸ Welfare across ϕ_r ▸ Other rules ▸ Reverse shock

◂ Back

Mechanism: Cost-Push Shock

solid: goods only dotted: $\phi_r^* = -0.48$

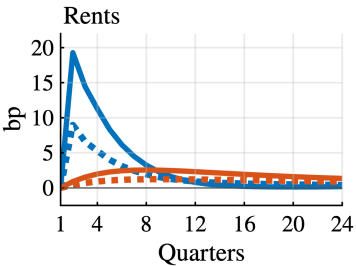


$\rightarrow$ lower landlord mortgage costs and savings returns

Back

Mechanism: Cost-Push Shock

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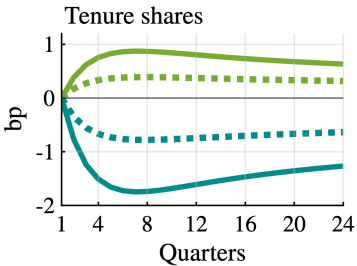
→ lower landlord mortgage costs and savings returns

→ rents rise less market average

Back

Mechanism: Cost-Push Shock

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→ lower landlord mortgage costs and savings returns

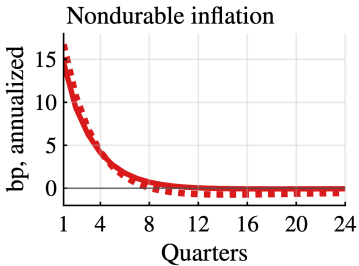
→ rents rise less

→ fewer renters renters owners

Back

Mechanism: Cost-Push Shock

solid: goods only dotted: $\phi_r^* = -0.48$



→ lower landlord mortgage costs and savings returns

→ rents rise less

→ fewer renters

→ some price destabilization

Back

Taking Stock

1. **Ease**: rent inflation should *lower* the policy rate
2. **Robust**: any $\phi_r \in [-0.25, 0]$ raises welfare
3. Lower rates $\rightarrow$ **landlords'** costs $\downarrow$ $\rightarrow$ rents rise less & fewer **renters**

$\Rightarrow$ **Neither tighten nor look through: ease**

Back

Ramsey Policy

Standard Ramsey Policy

chosen at the date of the shock

Problem $\max \ \mathcal{W}(\mathbf{i})$ s.t. competitive equilibrium given $\mathbf{i} = \{i_t\}_t$

Computational method automatic differentiation (Barcons, Ghassibe, Nakov & Thaler 2026)

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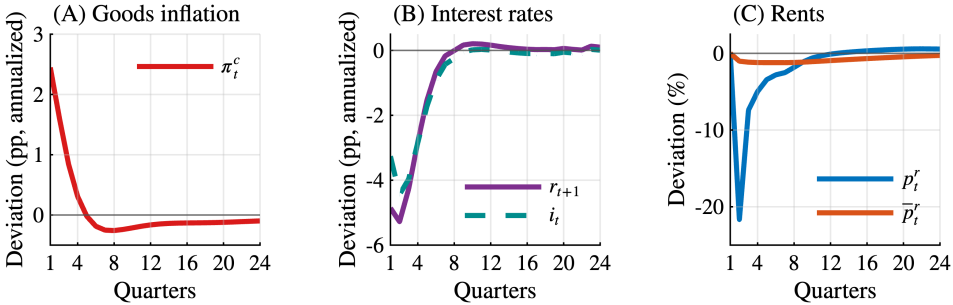
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- **No shock:** eases by surprise → redistribution to poor **renters** & **mortgagors**



Back

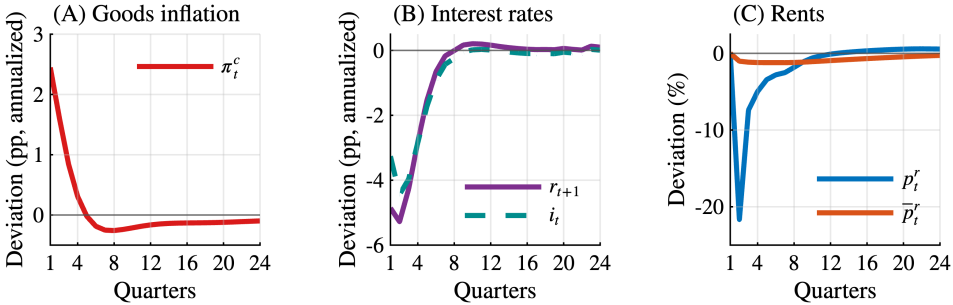
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- **No shock:** eases by surprise → redistribution to poor **renters** & **mortgagors**



- **Patience shock:** same surprise easing, about the same gain ▸ Results

◂ Back

Timeless Ramsey Policy

Timeless problem

What is optimal if the central bank cannot exploit the surprise?

Back

Timeless Ramsey Policy

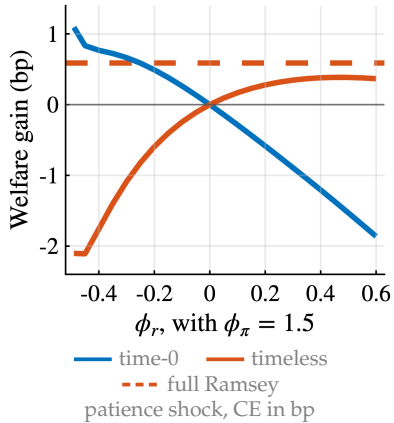
Timeless problem

What is optimal if the central bank cannot exploit the surprise?

Result: timeless is mostly stabilization

→ **no shock**: a surprise brings no gain

→ **patience**: best rule tightens, $\phi_\pi = 2.1$, $\phi_r = +0.2$



Back

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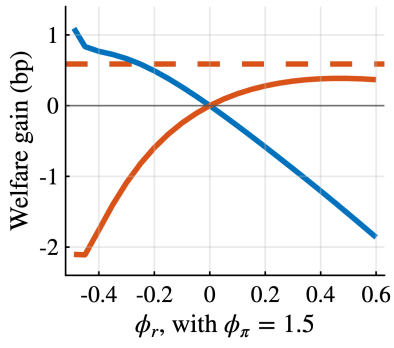
→ **patience**: best rule tightens, $\phi_\pi = 2.1$, $\phi_r = +0.2$

Why? The weights

Weights

→ **time-0**: easing helps *renters* & *borrowers*, who count **most**

→ **timeless**: promises to past *savers* & *landlords* weaken it



Back

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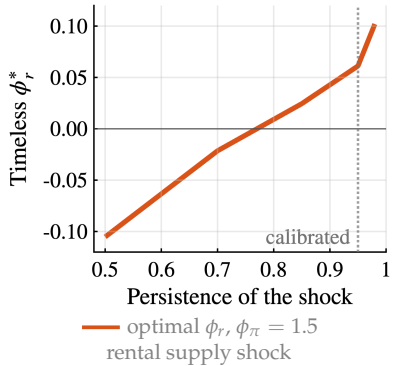
Weights

→ **time-0**: easing helps *renters* & *borrowers*, who count **most**
→ **timeless**: promises to past *savers* & *landlords* weaken it

It depends on shock persistence

Persistence

→ persistent: stabilization ➜ $\phi_r^* > 0$
→ transitory: redistribution ➜ $\phi_r^* < 0$

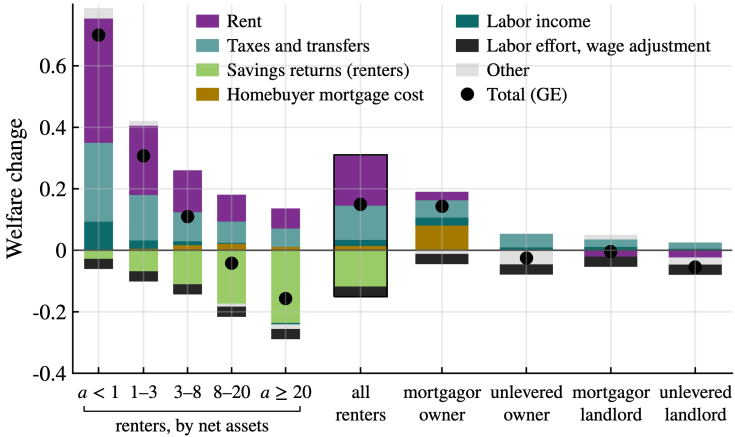


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Who Gains

Welfare change: $\phi_r^* = -0.50$ *vs* goods only (*in utils, per capita*)

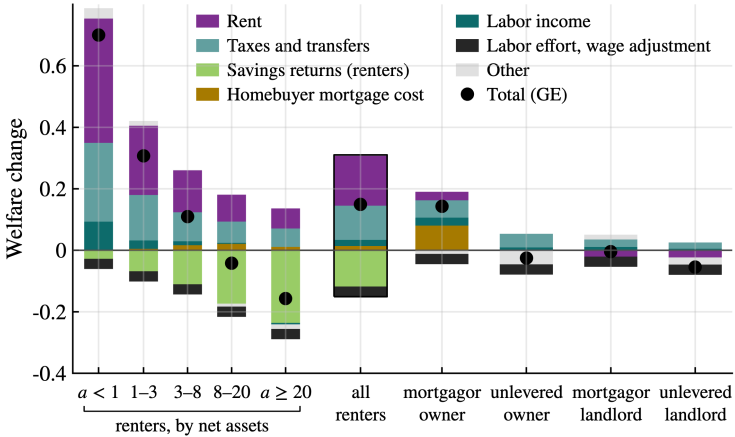
preferences shock, by household at the shock



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Who Gains

Welfare change: $\phi_r^* = -0.50$ *vs* goods only (*in utils, per capita*)
preferences shock, by household at the shock



Rent relief is progressive

→ lower rents: every renter gains, poorest most

→ wealthy renters lose: lower savings returns

→ mortgagor-owners gain: lower mortgage costs

→ lower interest bill: higher transfers

▸ Other shocks

◂ Back

Just a Less Hawkish Rule?

$\pi_t^{\text{rent}} \approx \pi_t^c + \text{real rent growth}$

$$i_t = r_{ss} \quad + \quad \underbrace{(\phi_\pi + \phi_r)\,\pi_t^c}_{\text{less hawkish on } \pi^c} \quad + \quad \underbrace{\phi_r\left(\pi_t^{\text{rent}} - \pi_t^c\right)}_{\text{eases with real rent growth}}$$

Is the gain just the first term?

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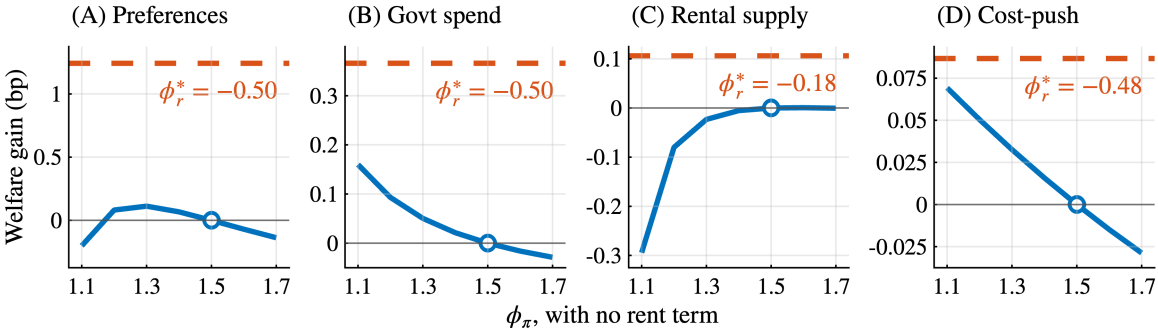
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Lowering ϕ_π alone falls short: the rent term matters

▸ Fixed hawkishness ▸ One rule, all shocks



no ϕ_π comes close | $\phi_\pi = 1.1$: $\approx$ half short | no ϕ_π comes close | $\phi_\pi = 1.1$: $\approx 1/5$ short

no rent term, varying ϕ_π — optimized rent rule ($\phi_\pi = 1.5$) — goods only — CE in bp

◂ Back

Appendix

VAR Details

Other countries

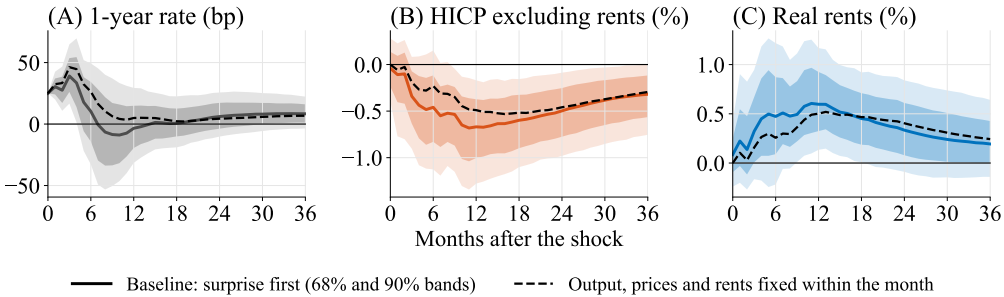
$$y_t = c + \sum_{\ell=1}^{6} A_\ell \, y_{t-\ell} + u_t, \qquad y_t = \left(z_t,\ i_t,\ ip_t,\ p_t,\ q_t,\ s_t\right)', \qquad u_t = B\,\varepsilon_t$$

	Variable	Role	As in
z_t	ECB surprise	instrument, ordered first	Altavilla et al. (2019); GK
i_t	1-year rate	policy indicator	Gertler and Karadi (2015)
ip_t	Industrial production	activity	GK; Dias and Duarte (2019)
p_t	HICP excluding rents	prices	GK; Dias and Duarte (2019)
q_t	HICP rents	rents	Dias and Duarte (2019)
s_t	ES–DE 10-year spread	credit conditions	GK's excess bond premium

- **Shock**: innovation in z_t (1-year OIS change around ECB announcements, monthly as in GK), ordered first, Cholesky (Plagborg-Møller and Wolf 2021); +25 bp on i_t; first-stage $F = 11.9$
- **Estimation**: Spain, monthly, 2005:3–2019:12; 6 lags; log levels; block bootstrap, 68% and 90% bands

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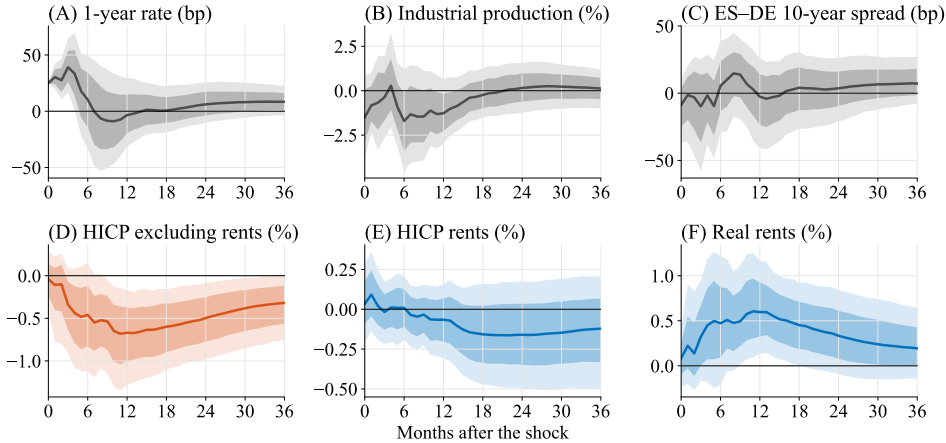
VAR: Robustness



- **Ordering**: surprise placed after output, prices and rents (dashed): prices -0.5%, real rents $+0.5\%$ after a year
- **INE rents** instead of HICP rents: real rents $+0.4\%$ after a year
- Real rents higher after a year in **all 24 specifications**: 6 or 12 lags, HICP or HICP excluding rents, two rent series, three sets of controls (none, spread, spread + oil)

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VAR: All Responses



Baseline: surprise ordered first, Spain 2005:3–2019:12, per 25 bp on the 1-year rate; median, 68% and 90% bands.

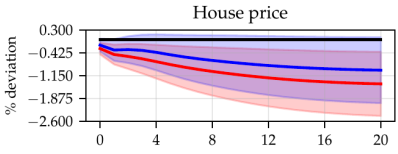
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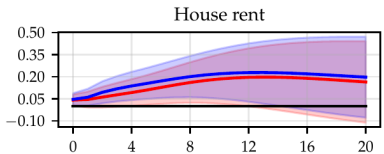
Evidence: Rents after a Tightening

United States

house prices fall,

index rents rise





68% bands. US: Dias and Duarte (2022), 25bp, quarters; rents = CPI rent of primary residence (all tenants).

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Euro area

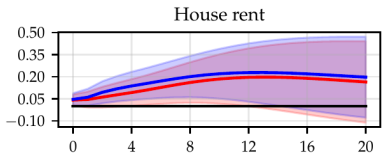
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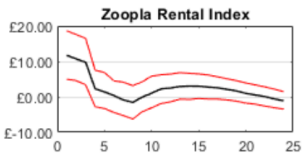


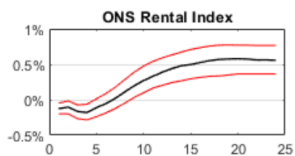


United Kingdom

new leases jump,

index rises later





68% bands. US: Dias and Duarte (2022), 25bp, quarters; rents = CPI rent of primary residence (all tenants). UK: Lazarowicz and Richard (2023), 100bp, months; Zoopla = new-lease listings (£/week, real), ONS = official private rents index.

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Euro area

Evidence: Euro Area

*"Data on new rents in the euro area are **scarce**."*

*"[T]he rent basket mostly covers **existing rents**, which tend to catch up with market forces **with a lag**."*

— ECB Economic Bulletin, Issue 7/2023, box "Rent inflation in the euro area"

Evidence: Euro Area

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Euro area

HICP actual rentals





→ house prices fall; index rents rise slowly, but the 68% band includes zero

→ as for the UK index: mostly existing leases, so it moves later and less than new leases

Corsetti et al. (2022): 25bp contractionary shock, quarters, 68% bands.

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Textbook Economy: Setup

Reduced **Iacoviello (2005)**: one household, owned housing in fixed supply

Household

$$\max \; E_0 \sum_t \beta^t \left[\frac{c_t^{1-\sigma}}{1-\sigma} + j \ln h_t \right]$$

$$\text{s.t.} \;\; c_t + q_t \left(h_t - h_{t-1} \right) + b_t = y_t + \left(1 + r_{t-1} \right) b_{t-1}, \qquad h_t = \overline{H}$$

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First-order conditions

$$c_t^{-\sigma} = \beta \left(1 + r_t \right) E_t \, c_{t+1}^{-\sigma} \qquad \text{(Euler)}$$

$$q_t \, c_t^{-\sigma} = \frac{j}{\overline{H}} + \beta \, E_t \left[q_{t+1} \, c_{t+1}^{-\sigma} \right] \qquad \text{(Housing demand: buy, enjoy, resell)}$$

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Rent no rental market in Iacoviello: R_t is the **shadow rent**, what one more period of housing services is worth (a frictionless rental market would charge exactly this; see Kaplan et al., 2020)

$$R_t \equiv \frac{j / \overline{H}}{c_t^{-\sigma}} \qquad \Rightarrow \qquad \underbrace{q_t}_{\text{buy today}} = \underbrace{R_t}_{\text{rent}} + \underbrace{\beta \, E_t \left[\left(\frac{c_{t+1}}{c_t} \right)^{-\sigma} q_{t+1} \right]}_{\text{resale value, discounted}}$$

Back

Textbook Economy: Response to a Tightening

Rent hats: log deviations from steady state

$$\hat{R}_t = \sigma\, \hat{c}_t, \qquad \hat{c}_t = E_t \hat{c}_{t+1} - \frac{1}{\sigma}\, \hat{r}_t \quad \Rightarrow \quad \hat{R}_t \;=\; E_t\, \hat{R}_{t+1} - \hat{r}_t \;=\; -\sum_{k \geq 0} E_t\, \hat{r}_{t+k}$$

a tightening raises the path of real rates $\Rightarrow$ **rents fall**, one-for-one

Textbook Economy: Response to a Tightening

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a tightening raises the path of real rates $\Rightarrow$ **rents fall**, one-for-one

House price

$$\text{Housing demand: } q_t\,c_t^{-\sigma} = \tfrac{j/\overline{H}}{1-\beta} \text{ (constant)} \quad \Rightarrow \quad q_t = \frac{R_t}{1-\beta} \quad \Rightarrow \quad \hat{q}_t = \hat{R}_t$$

rents and prices fall by the same %: **rent-to-price ratio unchanged** (data: it rises)

Textbook Economy: Response to a Tightening

Rent hats: log deviations from steady state

$$\hat{R}_t = \sigma\,\hat{c}_t, \qquad \hat{c}_t = E_t\hat{c}_{t+1} - \tfrac{1}{\sigma}\,\hat{r}_t \quad \Rightarrow \quad \hat{R}_t \;=\; E_t\,\hat{R}_{t+1} - \hat{r}_t \;=\; -\textstyle\sum_{k\geq 0} E_t\,\hat{r}_{t+k}$$

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rents and prices fall by the same %: **rent-to-price ratio unchanged** (data: it rises)

Missing: no rental market, so nobody is priced out of owning into renting, and no landlords whose mortgage costs rise with the policy rate

Back

Who Are the Landlords?

	No landlord textbook	**A separate sector** two-agent models: Dias & Duarte 2022, de Llanos 2026	**Households** this paper
Rent equals	what households will pay for a fixed stock	landlord's user cost + conversion wedge	where renters who cannot buy meet landlords who could lend instead
Demand	falls: spending falls	**shifts out**: buyers priced out	**shifts out**: buyers priced out
Supply	fixed stock	**falls** with house prices; steep because converting homes is costly	**shifts in**: letting pays less than lending, mortgages cost more
Rents	**fall**	**rise** only if the demand shift beats the cheaper home steepness = conversion cost, param.	**rise**; size of each shift = households at the buy/rent and own/let margins disciplined by tenure flows

This paper: both shifts; landlords supply about two thirds of the pressure from higher rates, and who switches (leverage, not tenure) drives incidence and welfare.

Back

Recursive Household Problem

State (a, h, y, s): wealth, tenure, productivity, lease state $s \in \{0, 1, \ldots, J\}$

housing services: $f(h) = \mathbb{1}_{h=0} + \mathbb{1}_{h>0} \left(\theta \tilde{h}_2 / \tilde{h}_1\right)^{\alpha_h}$

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Bellman equation

$$V_t^{\varepsilon}(a, h, y, s) \;=\; \max_{c, a', h'} \;\; \mathcal{U}(c, h, n) \;+\; \sigma_{\varepsilon}\, \varepsilon(h') \;+\; \beta_t\, \mathbb{E}_t \left[V_{t+1}(a', h', y', s') \,\middle|\, y, h, s, h' \right]$$

s.t. the budget and borrowing constraints

lease next period: keep rental tenure ($h' = h \in \{0, 2\}$) $\Rightarrow$ continues w.p. $\varphi\, \xi^s_{2-h,t}$, $s' = \min\{s+1, J\}$

otherwise $\Rightarrow$ new lease at the market rent ($s' = 0$) $\quad \xi$: partner's tenure retention

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Logit tenure choice $\widetilde{V}_t(h') \equiv \widetilde{V}_t(a, h, y, s; h')$: value of choosing h', before taste shocks

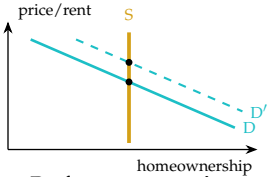
$$\mathbb{P}_t(h' \mid a, h, y, s) = \frac{\exp\left(\widetilde{V}_t(h') / \sigma_\varepsilon\right)}{\sum_j \exp\left(\widetilde{V}_t(j) / \sigma_\varepsilon\right)}, \qquad V_t(a, h, y, s) = \sigma_\varepsilon \log \sum_j \exp\left(\widetilde{V}_t(j) / \sigma_\varepsilon\right)$$

taste shocks ε (type-I extreme value; Iskhakov et al. 2017) only smooth the tenure choice

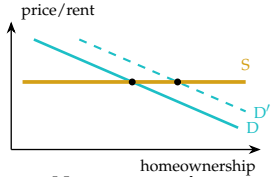
Back

Rental Supply across Models

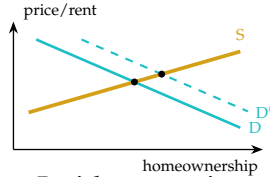
Tenure supply landlords' sales to owner-occupiers as price/rent rises (Greenwald & Guren 2025)



Perfect segmentation
Favilukis et al. (2017)
no rental market: credit moves only price/rent



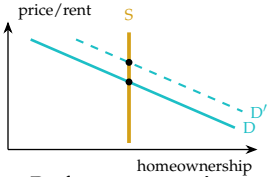
No segmentation
Kaplan et al. (2020)
deep-pocketed landlords: credit moves only homeownership



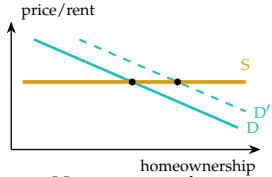
Partial segmentation
Greenwald & Guren (2025)
credit moves both; data closer to full segmentation

Rental Supply across Models

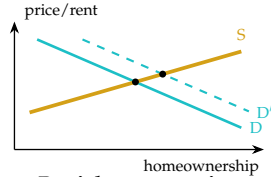
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credit moves both; data closer to full segmentation

This paper: upward-sloping, from the endogenous landlord margin (Castellanos et al. 2024)

rental-supply elasticity	model	evidence
quarters 1–4	0.37	0.4–0.7 (short run)
quarters 5–8	1.06	$\approx$ 1.4 (long run)

Back

Housing Costs and Collateral

Housing units sizes $\tilde{h}_1 < \tilde{h}_2$, freely convertible: a unit of size $\tilde{h}$ is worth $p_t^h \tilde{h}$

$\rightarrow$ renter holds nothing $\rightarrow$ owner holds $\tilde{h}_2$ $\rightarrow$ landlord holds $\tilde{h}_1 + \tilde{h}_2$

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from h \ to h'	renter (0)	owner (1)	landlord (2)
renter (0)	0	$(1+\tau_h)\,\tilde{h}_2$	$(1+\tau_h)(\tilde{h}_1+\tilde{h}_2)$
owner (1)	$-(1-\tau_h-\delta_h)\,\tilde{h}_2$	$\delta_h\,\tilde{h}_2$	$(1+\tau_h)\,\tilde{h}_1+\delta_h\,\tilde{h}_2$
landlord (2)	$-(1-\tau_h-\delta_h)(\tilde{h}_1+\tilde{h}_2)$	$\delta_h\,\tilde{h}_2-(1-\tau_h-\delta_h)\,\tilde{h}_1$	$\delta_h\,(\tilde{h}_1+\tilde{h}_2)$

$\rightarrow$ positive: net cost, negative: net receipt $\rightarrow$ τ_h: transaction tax $\rightarrow$ δ_h: depreciation

e.g. owner $\rightarrow$ renter: sells $\tilde{h}_2$ and receives $(1-\tau_h-\delta_h)\,p_t^h\,\tilde{h}_2$

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Housing bought $q(h, h')$ (sets the LTV limit)

$\rightarrow q(0,1) = \tilde{h}_2$ $\rightarrow q(0,2) = \tilde{h}_1 + \tilde{h}_2$ $\rightarrow q(1,2) = \tilde{h}_1$ $\rightarrow q = 0$ otherwise

Back

Average Rent and the Rent Index

Average rent paid by renters $\mu_t^{0,s}/\mu_t^0$: share of renters with lease state s

$$\overline{p}_t^r \;=\; \sum_{s<J}(\mu_t^{0,s}/\mu_t^0)\,p_{t-s}^r \;+\; (\mu_t^{0,J}/\mu_t^0)\,\widetilde{p}_t^r \;\approx\; \sum_{k\geq 0}\omega_k\,p_{t-k}^r$$

→ lease masses: $\mu_{t+1}^{0,s+1} = \varphi\,\xi_{0,t}^s\,\xi_{2,t}^s\,\mu_t^{0,s}$

→ long-term tenants: $\widetilde{p}_t^r = \lambda_{t-1}\,\widetilde{p}_{t-1}^r + (1-\lambda_{t-1})\,p_{t-J}^r$

→ weights ω_k on the market rent k quarters ago: 59% on leases younger than J, 41% on long-term tenants

Rent inflation in the index: $1+\pi_t^{\text{rent}} = (1+\pi_t^c)\,\overline{p}_t^r/\overline{p}_{t-1}^r$

→ mean lag of 22 quarters: turnover delays the market rent, without shrinking it

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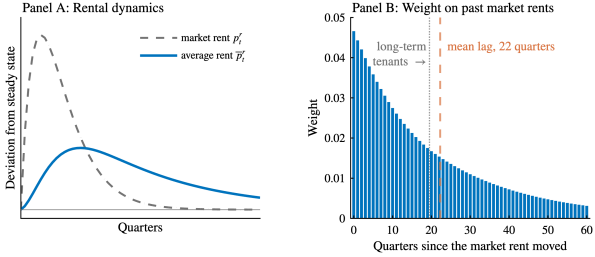
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Back

Alternative Taylor Rules

Baseline $i_t = r_{ss} + \phi_\pi \, \pi_t^{\text{hicp}} + \varepsilon_t^m$

Inertia $i_t - r_{ss} = \rho_i \, (i_{t-1} - r_{ss}) + (1 - \rho_i) \, \phi_\pi \, \pi_t^{\text{hicp}} + \varepsilon_t^m$

$\rho_i = 0.8$ (Burriel et al. 2010; Smets & Wouters 2007); one-time shock, same size as baseline

Output gap $i_t = r_{ss} + \phi_\pi \, \pi_t^{\text{hicp}} + \phi_y \left(Y_t^c / Y_{ss}^c - 1 \right) + \varepsilon_t^m$

$\phi_y = 0.125$, i.e. 0.5 annualized (Burriel et al. 2010)

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Results hold; only magnitudes change ▸ Inertia: IRFs ▸ Output gap: IRFs

→ rents still rise while house prices fall; landlords still drive about 65% of the rise

→ lower rent peak: −15% with inertia, −13% with the output gap

inertia: per unit of real rate, which moves 61% more; output gap: the real rate moves 14% less

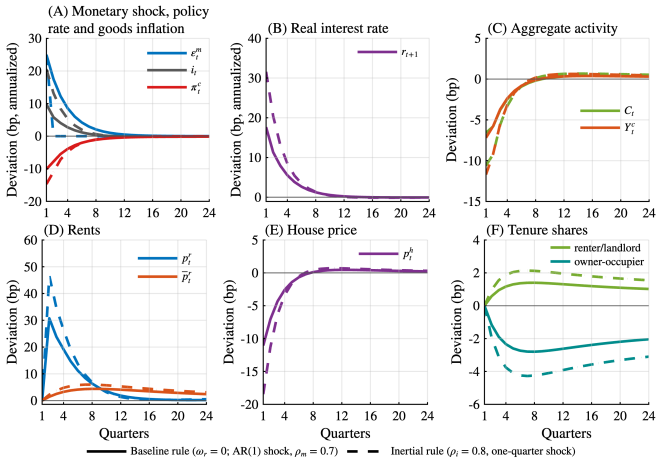
→ rents in the target still deepen the contraction, by less

→ optimal response: still ease, under all four shocks

◂ Back: model ◂ Back: optimal weight

Inertial Rule: IRFs

Figure: inflation term goods only ($\omega_r = 0$)

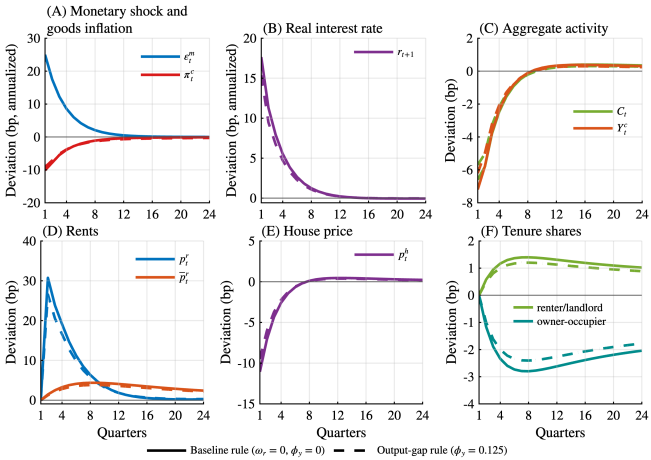


$\rightarrow$ one-time shock, same size as baseline

Back

Output-Gap Rule: IRFs

Figure: inflation term goods only ($\omega_r = 0$)



→ same signs and shapes, but smaller

→ why? rule cuts rates as output falls

Back

A Spanish Central Bank?

The ECB's rule responds to euro-area inflation; Spain's share $\vartheta = 11.4\%$

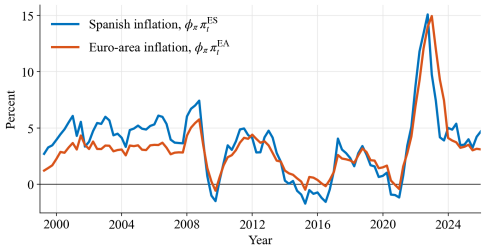
$$i_t = r_{ss} + \phi_\pi \left[\vartheta \, \pi_t^{\text{ES}} + (1 - \vartheta) \, \pi_t^{\text{xES}} \right] + \varepsilon_t^m$$

Seen from Spain

$\rightarrow$ alone, the response to Spanish inflation is small: $\phi_\pi \vartheta \approx 0.17$

$\rightarrow$ but inflation in Spain and in the rest of the euro area shares one cycle: **correlation 0.82**, slope $b = 0.75$

$\rightarrow$ effective coefficient $\phi_\pi \left[\vartheta + (1 - \vartheta) \, b \right] =$ **1.16**



$\phi_\pi \pi_t$, $\phi_\pi = 1.5$, y-o-y HICP 1999–2025; corr. 0.86

$\Rightarrow$ **to first order, a fictitious Spanish central bank acts as the ECB**

Back

Wage Setting

Unions a continuum $k \in [0,1]$; union k sets its nominal wage W_{kt}

Labor demand $N_{kt} = (W_{kt}/W_t)^{-\epsilon_t} N_t$ (a packer bundles the varieties)

Rationing every household works the same hours: $n_{it} = N_t$

Objective $\sum_{j \geq t} D_{t,j} \Big[\int \left(u(c_{ij}, h_{ij}) - v(n_{ij}) \right) di - \frac{\theta_w}{2} \left(\pi^w_{kj} \right)^2 \Big]$ $D_{t,j}$: discount, θ_w: Rotemberg cost

Wage Phillips curve symmetric equilibrium, $W_{kt} = W_t$

$$\pi^w_t = \frac{\epsilon_t}{\theta_w} N_t \Big(v'(N_t) - (1+\tau_L) \tfrac{\epsilon_t - 1}{\epsilon_t} (1-\tau_y) w_t \int y_{it} u_c(c_{it}, h_{it}) di \Big) + \beta_t \pi^w_{t+1}$$

$v'(N_t)$: what an extra hour **costs**

$(1-\tau_y) w_t \int y_{it} u_c di$: what an extra hour **pays**, after tax

$(1+\tau_L) \frac{\epsilon_t - 1}{\epsilon_t}$: unions' desired markup, offset by the subsidy in SS

$\rightarrow$ unions push wages up when an hour **costs** more than it **pays**

Back

Competitive Equilibrium

Given the shock paths, an equilibrium is prices $(w_t, r_t, i_t, \pi_t^c, p_t^h, p_t^r, \widetilde{p}_t^r)$, quantities, transfers T_t, debt B_t, retention rates ξ_t and a distribution $g_t(a, h, y, s)$ such that:

Agents optimize

households: Bellman equation s.t. constraints

goods firm: $w_t = A^c$

construction firm: $Y_t^h = A^h L_t^{\alpha_L} S_t^{1-\alpha_L}$

maximizes profits given p_t^h

unions: $n_t = N_t$, wage Phillips curve

Markets clear

housing: $H_{t+1} = (1 - \delta_h) H_t + Y_t^h$

rental: renters = landlords

bonds: $A_t \equiv \int a \, dg_t = B_t$

land: $L_t = \overline{L}$

goods: by Walras' law

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Policy and aggregation

central bank: Taylor rule; Fisher: $1 + i_t = (1 + r_{t+1})(1 + \pi_{t+1}^c)$

government: debt absorbs deficits; the transfer responds to debt, $T_t = T_{ss} - \Gamma_B \left(B_t - \overline{B}\right)$

leases: expected retention $\xi_{h,t}^s$ = the one choices generate; long-term tenants' rent $\widetilde{p}_t^r$ follows its rule

g_t evolves with household choices, income shocks and lease resets

no aggregate risk: one-time unexpected (MIT) shocks, then a deterministic transition back to steady state

Back

Calibrated Parameters

Set externally

Housing and tenure

$\tilde{h}_1$, $\tilde{h}_2$	0.905, 1.095	housing sizes
α_h	0.5	housing preference
φ	0.99	lease continues, per quarter
J	20	quarters until long-term tenant

Mortgages

λ_{LTV}	0.8	20% down payment (Bover et al. 2019)
λ_{LTI}	24	6× annual income (IMF 2026)
κ	0.00345	1.38 pp spread, as in the data
m	0.0190	90% repaid in 30 years

Preferences and income

γ	2	risk aversion
η	2	inverse Frisch elasticity
ρ_y	0.95	income AR(1) (Flodén & Lindé 2001)
σ_y	0.104	innovation s.d. (Arellano et al. 2022)

$\tilde{h}_1$, $\tilde{h}_2$, α_h, α_L: Castellanos et al. (2024)

Production and policy

α_L	0.33	land share
ϵ	10	elasticity across labor varieties
τ_L	1/9	employment subsidy: offsets the markup
θ_w	777	wage PC slope 0.02 (Auclert et al. 2024)
ϕ_π	1.5	Taylor rule (Auclert et al. 2025)
$\overline{B}$	4	debt: 100% of annual GDP
Γ_B	0.051	fiscal rule (Burriel et al. 2010)

Set internally, jointly

A^h	0.0294	housing productivity
δ_h	0.0046	depreciation (1.8% a year)
θ	1.197	ownership preference
β	0.9881	discount factor
σ_ε	0.122	taste shocks
τ_h	0.025	transaction tax
τ_y	0.150	labor tax

normalized: $A^c = \overline{L} = \mathbb{E}[y] = 1$; ψ_n sets $N = 1$

Back

Additional Moments

	Model	Spain
Renters		
renters' labor income / mean	0.83	0.74
rent burden, median	30.7%	24.1%
rent share of the consumption basket	5.3%	3.1%
Wealth		
top-10% share of net worth	18.1%	53.2%
wealthy hand-to-mouth	16.0%	15.2%
poor hand-to-mouth	1.7%	4.4%
Investment		
nonresidential investment / GDP	—	15.6%

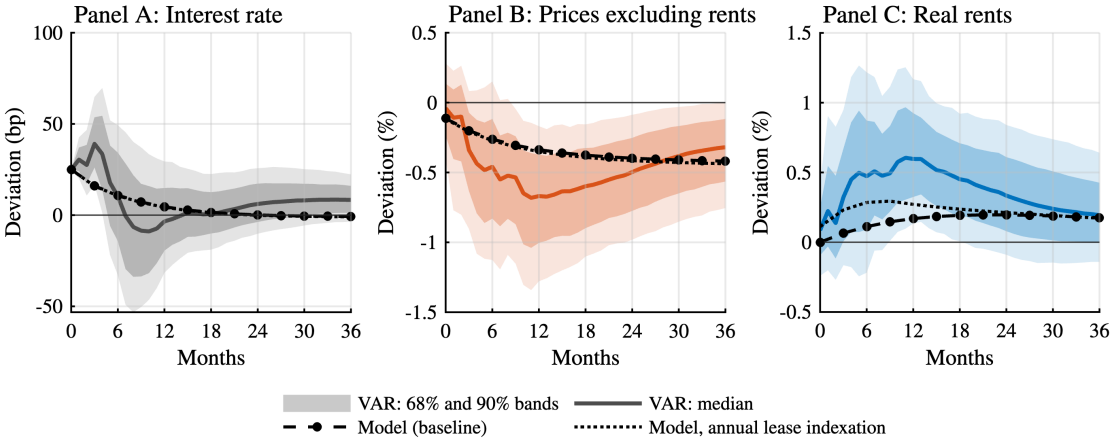
→ common discount factor: thin top tail (Auclert et al. 2025: heterogeneous patience)

→ renters cannot borrow: few poor hand-to-mouth; the ordering is right

→ adding productive capital: another saving vehicle

Back

Monetary Tightening in Spain: Model and Evidence



→ prices fall, real rents rise: same sign and timing as the VAR

Model scaled: 1-year nominal yield +25 bp on impact. Dotted: annual lease indexation.

Back

Costs across Rent Weights



Cumulative, quarters 1–8 and 1–20: % difference from goods only ($\omega_r = 0$); negative: lower

→ costs grow almost in proportion to the weight; the disinflation does not

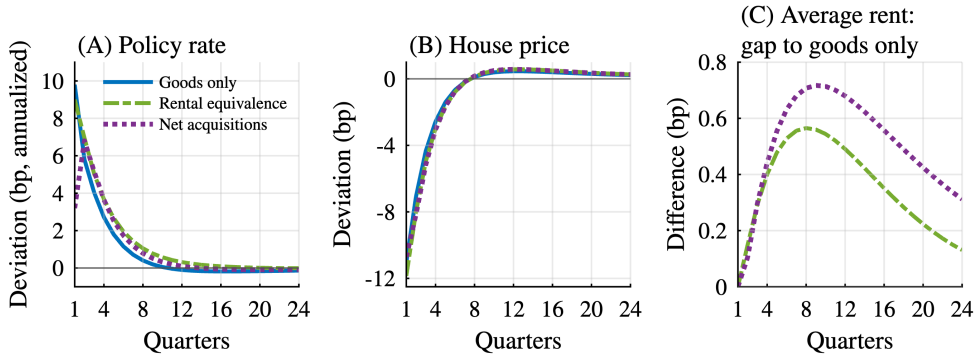
→ at Spain's 3.1%: about 60% of the effects at the model's 5.3%

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Owner-Occupied Housing in the Target

Index $\pi_t^{\text{index}} = (1-\omega_r-\omega_o)\,\pi_t^c + \omega_r\,\pi_t^{\text{rent}} + \omega_o\,\pi_t^o$ $\quad \omega_r = 5.3\% \times (1-\omega_o)$

Rental equivalence $\pi_t^o = \pi_t^{\text{rent}}$ $(\omega_o = 13\%)$ $\quad$ **Net acquisitions** $1+\pi_t^o = (1+\pi_t^c)\,p_t^h/p_{t-1}^h$ $(\omega_o = 9\%)$



→ rental equivalence: imputed rents = average rent ⇒ the same lagged feedback, 3–4× stronger

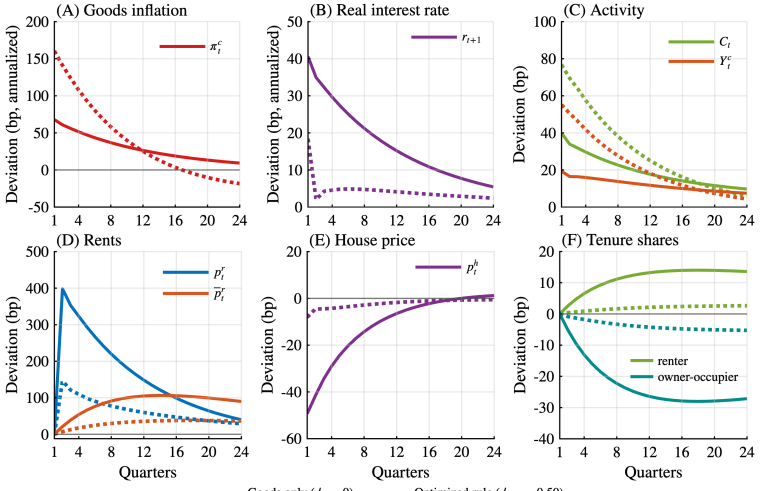
→ net acquisitions reads house-price *growth*: eases on impact, tightens as prices recover

the tightening comes in quarters 2–4, when new leases are signed at the peak market rent

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Preferences Shock: IRFs

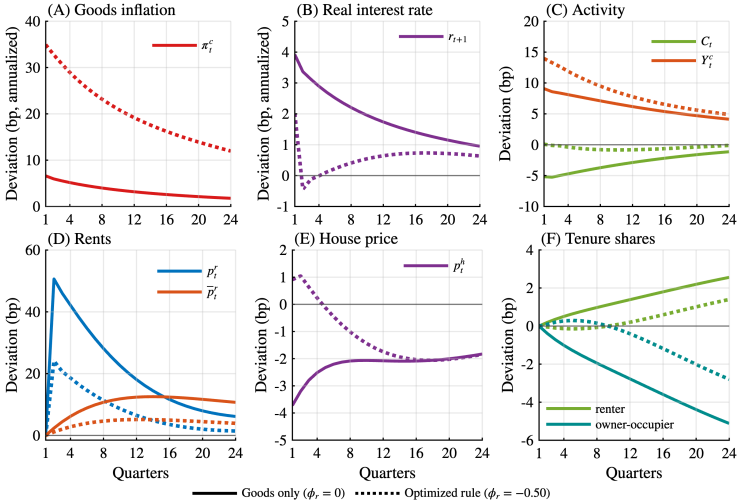
Figure: inflation term goods only ($\omega_r = 0$)



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Government Spending Shock: IRFs

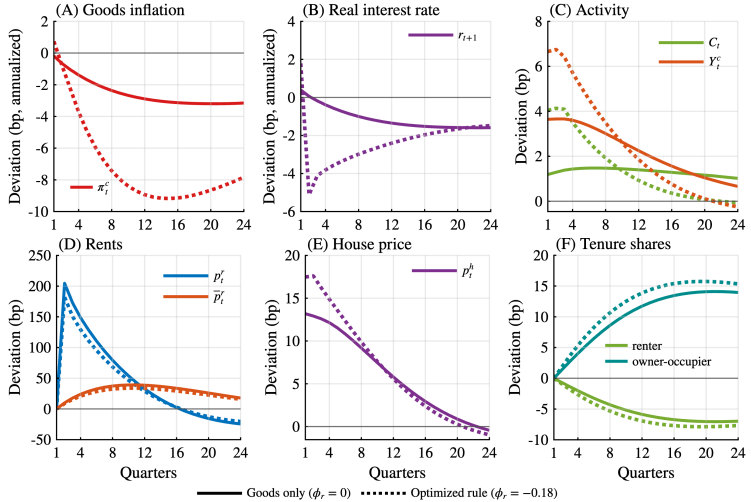
Figure: inflation term goods only ($\omega_r = 0$)



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Rental Supply Shock: IRFs

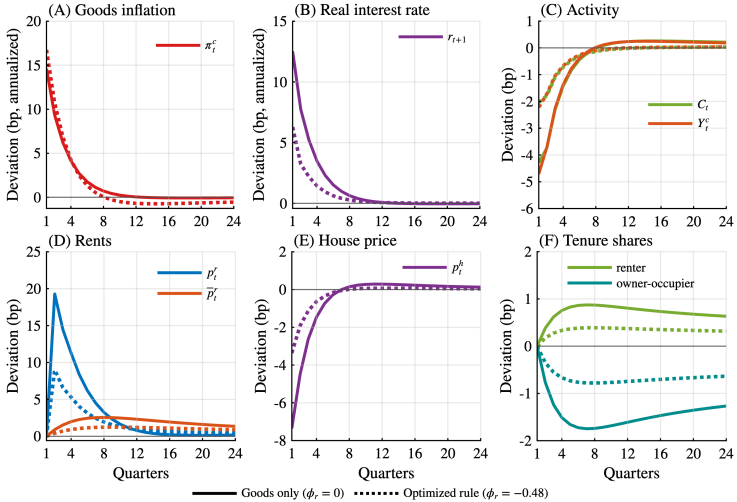
Figure: inflation term goods only ($\omega_r = 0$)



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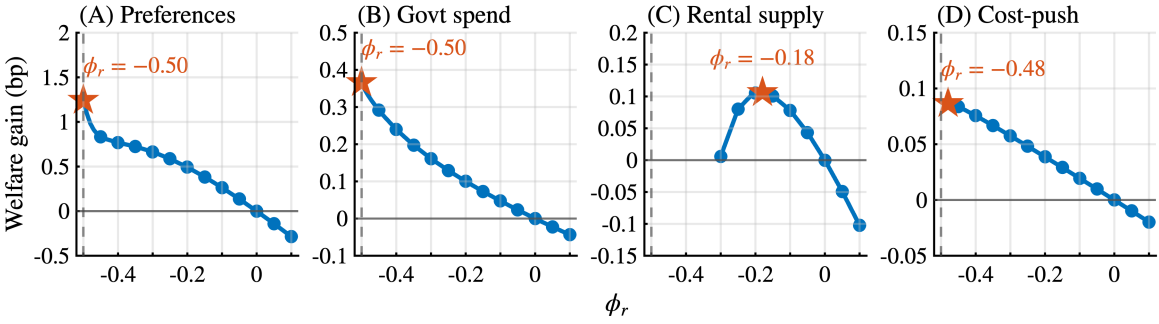
Cost-Push Shock: IRFs

Figure: inflation term goods only ($\omega_r = 0$)



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Welfare across ϕ_r



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Reverse Sign Shock

each shock with the opposite sign of the baseline
$\Rightarrow$ rents fall under all four

$\rightarrow$ **optimal $\phi_r > 0$ under all four shocks** ▸ IRFs ▸ Who gains
at the cap $\phi_r = +0.65$ under all four; gains $\geq +2.40, +0.24, +0.65, +0.14$ bp

$\rightarrow$ **why?** mostly redistribution, also efficiency *(smaller)*

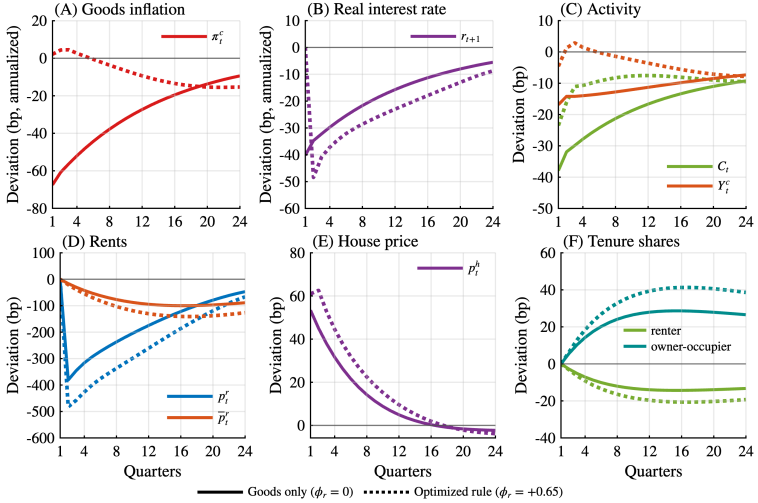
$\rightarrow$ rent term **loosens policy**: lower real rate, higher inflation, both signs

$\rightarrow$ redistribution always gains; efficiency gains only when the shock is deflationary

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Reverse Sign Shock: IRFs

Preferences shock; inflation term goods only ($\omega_r = 0$)

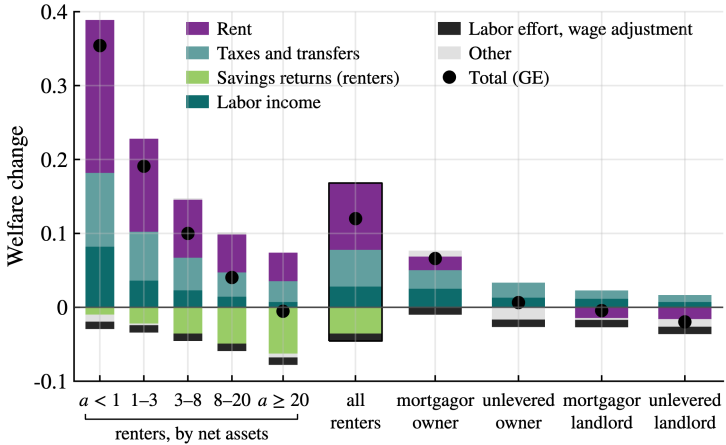


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Reverse Sign: Who Gains

Welfare change: $\phi_r = +0.65$ *vs* goods only (*in utils, per capita*)

preferences shock, reverse sign, by household at the shock



Rent relief leads

→ lower rents: every renter gains, poorest most

→ then higher transfers and labor income

→ wealthy renters lose: lower savings returns

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Holding Hawkishness Fixed

welfare gain over $(\phi_\pi, \phi_r) = (1.5, 0)$, bp

$$i_t = r_{ss} + \underbrace{(\phi_\pi + \phi_r)}_{\widetilde{\phi}_\pi\text{: hawkishness on } \pi^c} \pi_t^c + \phi_r \left(\pi_t^{\text{rent}} - \pi_t^c\right)$$

	Less hawkish only best $\widetilde{\phi}_\pi$, $\phi_r = 0$	**Rent term only** $\widetilde{\phi}_\pi = 1.5$, $\phi_r = -0.4$	**Both** best on the grid, (ϕ_π, ϕ_r)
Preferences	$+0.11$ ($\widetilde{\phi}_\pi = 1.3$)	$+\mathbf{0.58}$	$+0.78$ (1.6, -0.4)
Govt spend	$+\mathbf{0.16}$ ($\widetilde{\phi}_\pi = 1.1$)	$+0.09$	$+0.24$ (1.5, -0.4)
Rental supply	0.00 ($\widetilde{\phi}_\pi = 1.5$)	$+\mathbf{0.28}$	$+0.28$ (1.9, -0.4)
Cost-push	$+\mathbf{0.07}$ ($\widetilde{\phi}_\pi = 1.1$)	$+0.02$	$+0.08$ (1.5, -0.4)

→ at a fixed $\widetilde{\phi}_\pi$ (≥ 1.2), easing with real rent growth helps under all four shocks

→ rent term matters most: preferences, rental supply; lower $\widetilde{\phi}_\pi$: govt, cost-push

→ rental supply: lowering $\widetilde{\phi}_\pi$ alone *hurts* (-0.29 at $\widetilde{\phi}_\pi = 1.1$)

grid: $\widetilde{\phi}_\pi \in \{1.1, 1.2, 1.3, 1.5\}$, $\phi_r \in [-0.4, 0]$; best points at $\phi_r = -0.4$

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One Rule for All Shocks

$\pi_t^{\text{rent}} \approx \pi_t^c + \text{real rent growth}$

$$i_t = r_{ss} + \underbrace{(\phi_\pi + \phi_r)\,\pi_t^c}_{\text{response to } \pi^c\text{: 1.3 in both}} + \underbrace{\phi_r\left(\pi_t^{\text{rent}} - \pi_t^c\right)}_{\text{eases with real rent growth}}$$

Is the gain just the first term?

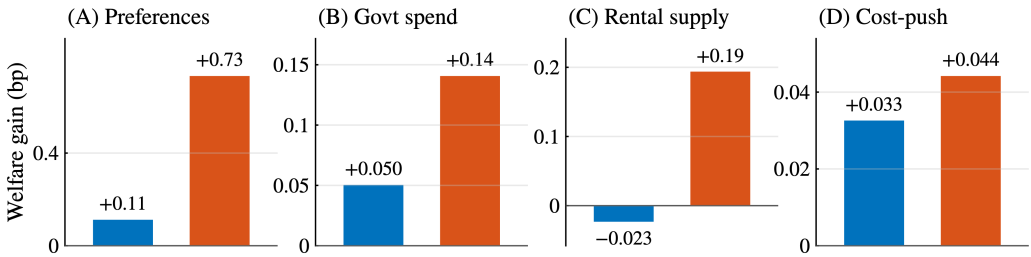
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One Rule for All Shocks

$\pi_t^{\text{rent}} \approx \pi_t^c +$ real rent growth

$$i_t = r_{ss} + \underbrace{(\phi_\pi + \phi_r)\,\pi_t^c}_{\text{response to } \pi^c\text{: 1.3 in both}} + \underbrace{\phi_r\left(\pi_t^{\text{rent}} - \pi_t^c\right)}_{\text{eases with real rent growth}}$$

Same response to π^c: higher welfare under every shock ▸ Fixed hawkishness



■ $(\phi_\pi, \phi_r) = (1.3, 0)$: best rule without rents ■ $(1.7, -0.40)$: best rule with rents best: summed over shocks · CE in bp

→ rent term **eases when rents rise**; no single ϕ_π does ▸ Same rates?

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Same Rates without a Rent Term?

cumulative real rate r_{t+1}, quarters 1–20

Each shock: rule without rents, same total rate cut as the rent rule → welfare?

	No rent term, same rates		**Rent term**
	ϕ_π that matches	its gain	$(1.7, -0.40)$
Preferences	1.11	-0.18	$+\mathbf{0.73}$
Govt spend	below 1.1	—	$+0.14$
Rental supply	none: rates fall	—	$+\mathbf{0.19}$
Cost-push	1.23	$+0.046$	$+0.044$

→ same rate path: a **different ϕ_π for every shock**

→ rental supply: the rent rule cuts rates; no ϕ_π does

→ cost-push: same rates, same gain → the gain is the lower rate path

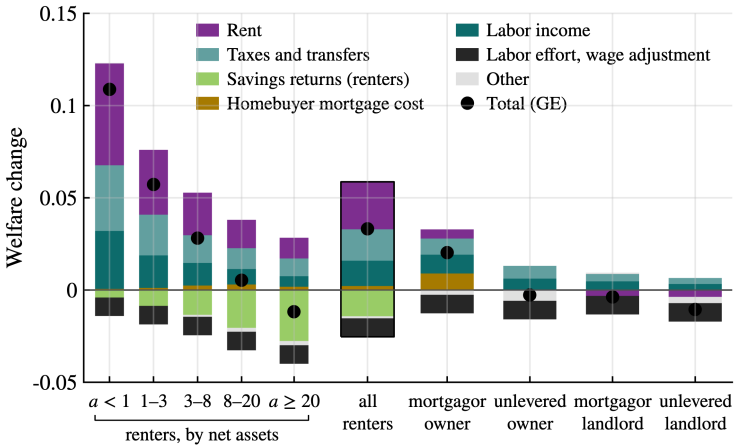
CE in bp over $(1.5, 0)$

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Who Gains: Other Shocks

Welfare change: $\phi_r^* = -0.50$ *vs* goods only (*in utils, per capita*)

govt spending shock, by household at the shock



Rent relief leads

→ lower rents: every renter gains, poorest most

→ more work: higher labor income, at an effort cost to all

→ lower interest bill: higher transfers

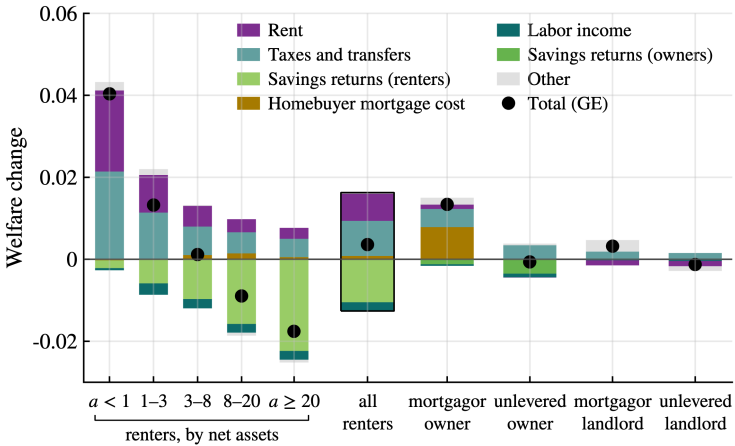
→ wealthy renters lose: lower savings returns

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Who Gains: Other Shocks

Welfare change: $\phi_r^* = -0.18$ *vs* goods only (*in utils, per capita*)

rental supply shock, by household at the shock



Transfers and rent relief

→ poorest renters: transfers and lower rents, about equally

→ wealthy renters lose: lower savings returns

→ renters as a group: about break even

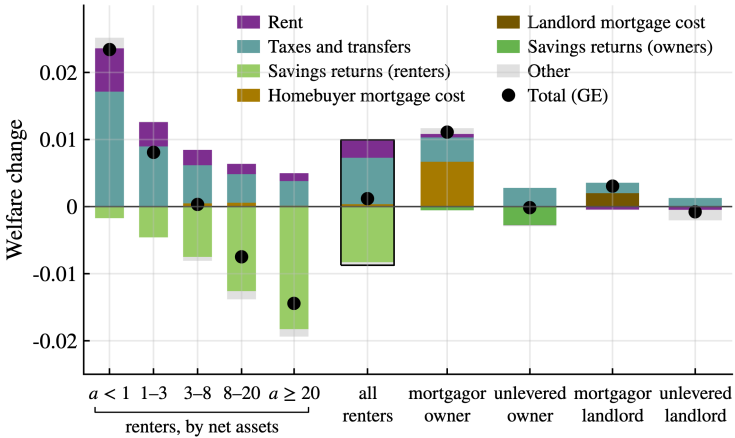
→ mortgagor-owners gain: lower mortgage costs

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Who Gains: Other Shocks

Welfare change: $\phi_r^* = -0.48$ *vs* goods only (*in utils, per capita*)

cost-push shock, by household at the shock



Transfers lead

→ poorest renters: mostly transfers, then lower rents

→ wealthy renters lose: lower savings returns

→ renters as a group: about break even

→ mortgagor-owners gain: lower mortgage costs

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Efficiency *vs* Redistribution

$dV_i^\lambda / d\phi_r$: household i's willingness to pay for a higher ϕ_r (*in* λ *units*)

$$d\mathcal{W}^\lambda / d\phi_r \quad = \quad \underbrace{\mathbb{E}_i\left[dV_i^\lambda / d\phi_r\right]}_{\textbf{efficiency}} \quad + \quad \underbrace{\mathbb{C}\text{ov}_i\left(\omega_i,\, dV_i^\lambda / d\phi_r\right)}_{\textbf{redistribution}}$$

efficiency: total willingness to pay — **redistribution**: toward high-ω_i households

ω_i: i's value of extra consumption, relative to the average household (Dávila & Schaab 2025) Decomposition

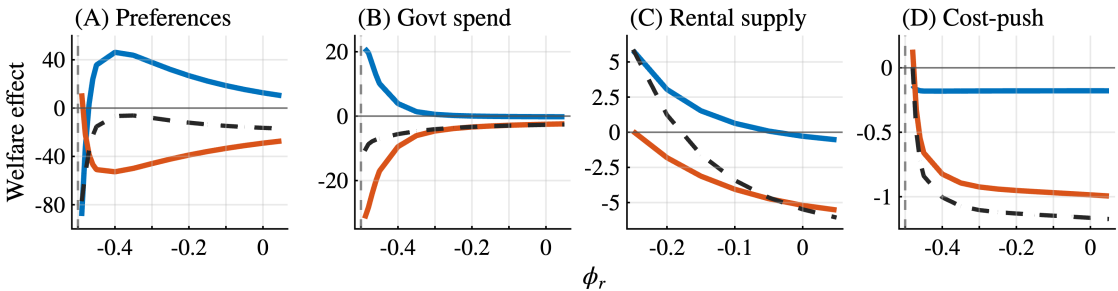
Why does easing raise welfare?

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Efficiency *vs* Redistribution

$$d\mathcal{W}^{\lambda}/d\phi_r \quad = \quad \underset{\text{total willingness to pay}}{\textbf{efficiency}} \quad + \quad \underset{\text{toward high-}\omega_i\text{ households}}{\textbf{redistribution}}$$

Mostly redistribution; under cost-push, efficiency too



Efficiency *vs* Redistribution: Decomposition

Welfare $\mathcal{W} = \sum_t \tilde{\beta}_t \big[\int u(c_{it}, h_{it})\, di - \nu(N_t) - \frac{\theta_w}{2} (\pi_t^w)^2 \big] = \sum_i \mu_i V_i$

$V_i = \mathbb{E}\big[\sum_t \tilde{\beta}_t \big(u(c_{it}, h_{it}) - \nu(N_t) - \frac{\theta_w}{2} (\pi_t^w)^2 \big) \big]$

i: household type at the shock (wealth, income, tenure), with share μ_i

Unit $\lambda_i = \mathbb{E}\big[\sum_t \tilde{\beta}_t\, u_c(c_{it}, h_{it}) \big]$: value of one unit of consumption forever

$\bar{\lambda} = \sum_i \mu_i \lambda_i, \qquad \omega_i = \lambda_i / \bar{\lambda}, \qquad dV_i^\lambda / d\phi_r = \lambda_i^{-1}\, dV_i / d\phi_r$

Split $d\mathcal{W}^\lambda / d\phi_r \equiv \bar{\lambda}^{-1}\, d\mathcal{W} / d\phi_r = \sum_i \mu_i\, \omega_i\, dV_i^\lambda / d\phi_r$

$$= \underbrace{\mathbb{E}_i \big[dV_i^\lambda / d\phi_r \big]}_{\textbf{efficiency}} + \underbrace{\mathbb{C}\text{ov}_i \big(\omega_i,\, dV_i^\lambda / d\phi_r \big)}_{\textbf{redistribution}}$$

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Efficiency *vs* Redistribution: Decomposition

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$$= \underbrace{\mathbb{E}_i \big[dV_i^\lambda / d\phi_r \big]}_{\textbf{efficiency}} + \underbrace{\mathbb{C}\text{ov}_i \big(\omega_i,\, dV_i^\lambda / d\phi_r \big)}_{\textbf{redistribution}}$$

→ **efficiency** > 0: on average, households would pay for a higher ϕ_r

→ **redistribution** > 0: a higher ϕ_r favors the households with high ω_i

→ ω_i, high to low: renters, then owners, then landlords; within each, mortgagors first

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Standard Ramsey: All Shocks

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Timeless Problem

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Welfare Weights

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Persistence of the Shock

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